

Chapter 21

Minimum Variance Estimators; Method of Moments

Lecture 21 | Devore 9e: 6.1–6.2

21.1 The sample variance is unbiased

We owe a proof from last lecture. First an algebraic identity.

Lemma 21.1.

$$\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n X_i^2 - n\bar{X}^2 = \sum_{i=1}^n X_i^2 - \frac{1}{n} \left(\sum_{i=1}^n X_i \right)^2.$$

Proof. Expand the square and use $\sum_i X_i = n\bar{X}$:

$$\sum_i (X_i - \bar{X})^2 = \sum_i X_i^2 - 2\bar{X} \sum_i X_i + n\bar{X}^2 = \sum_i X_i^2 - 2n\bar{X}^2 + n\bar{X}^2 = \sum_i X_i^2 - n\bar{X}^2. \quad \square$$

Theorem 21.2. If $X_1, \dots, X_n$ are i.i.d. with mean μ and variance σ^2 , then $\mathbb{E}[S^2] = \sigma^2$.

Proof. The two ingredients are $\mathbb{E}[Z^2] = \text{Var}(Z) + \mathbb{E}[Z]^2$ applied twice:

$$\mathbb{E}[X_i^2] = \sigma^2 + \mu^2, \quad \mathbb{E}\left[\left(\sum_i X_i\right)^2\right] = \text{Var}\left(\sum_i X_i\right) + \left(\mathbb{E}\left[\sum_i X_i\right]\right)^2 = n\sigma^2 + (n\mu)^2,$$

the middle step using independence. Then by the lemma,

$$\begin{aligned} \mathbb{E}[S^2] &= \frac{1}{n-1} \mathbb{E}\left[\sum_i X_i^2 - \frac{1}{n} (\sum_i X_i)^2\right] \\ &= \frac{1}{n-1} \left\{ n(\sigma^2 + \mu^2) - \frac{1}{n} (n\sigma^2 + n^2\mu^2) \right\} \\ &= \frac{1}{n-1} \left\{ n\sigma^2 + n\mu^2 - \sigma^2 - n\mu^2 \right\} = \frac{(n-1)\sigma^2}{n-1} = \sigma^2. \quad \square \end{aligned}$$

21.2 Which unbiased estimator is best?

Unbiasedness is a weak requirement: many estimators have it, and they can be very different. Recall the Uniform $[0, \theta]$ problem.

Example 21.3 (Two unbiased estimators of θ). $X_1, \dots, X_n$ i.i.d. Uniform $[0, \theta]$.

- From last lecture, $\hat{\theta}_1 = \frac{n+1}{n} \max(X_1, \dots, X_n)$ is unbiased.
- Also $\hat{\theta}_2 = 2\bar{X}$ is unbiased, since $\mathbb{E}[X_i] = \theta/2$ gives $\mathbb{E}[\hat{\theta}_2] = 2 \cdot \frac{\theta}{2} = \theta$.

Both hit the target on average. Now compare their variances. Write $M = \max(X_1, \dots, X_n)$, whose density $f_M(t) = nt^{n-1}/\theta^n$ we found last lecture. Then

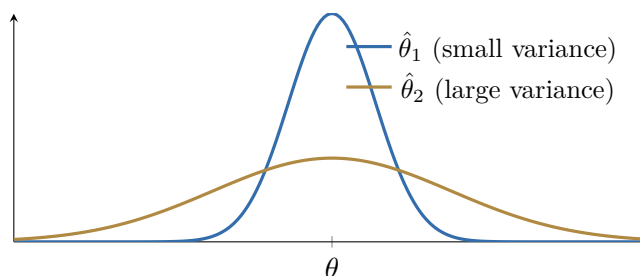
$$\mathbb{E}[M^2] = \int_0^\theta t^2 \cdot \frac{nt^{n-1}}{\theta^n} dt = \frac{n\theta^2}{n+2}, \quad \text{Var}(M) = \frac{n\theta^2}{n+2} - \left(\frac{n\theta}{n+1}\right)^2 = \frac{n\theta^2}{(n+2)(n+1)^2},$$

the last step because $(n+1)^2 - n(n+2) = 1$. Since $\hat{\theta}_1 = \frac{n+1}{n}M$,

$$\text{Var}(\hat{\theta}_1) = \left(\frac{n+1}{n}\right)^2 \text{Var}(M) = \frac{\theta^2}{n(n+2)}, \quad \text{Var}(\hat{\theta}_2) = \frac{4}{n} \text{Var}(X_i) = \frac{4}{n} \cdot \frac{\theta^2}{12} = \frac{\theta^2}{3n}.$$

Since $n(n+2) > 3n$ for $n > 1$,

$$\text{Var}(\hat{\theta}_1) < \text{Var}(\hat{\theta}_2) \quad \implies \quad \hat{\theta}_1 \text{ is the better estimator.}$$



Remark 21.4. The difference is dramatic: $\text{Var}(\hat{\theta}_1)$ shrinks like $1/n^2$ while $\text{Var}(\hat{\theta}_2)$ shrinks only like $1/n$. Using the maximum extracts far more information about θ than the average does — which makes sense, since only the largest observation carries information about where the interval ends.

Definition 21.5 (MVUE). Among all unbiased estimators of θ , the one with the smallest variance is the **minimum variance unbiased estimator** (MVUE).

Remark 21.6 (Is a biased estimator always worse?). No. An estimator with a small bias and a much smaller variance can beat an unbiased one in every practical sense. The standard way to trade the two off is the **mean squared error**

$$\text{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2] = \text{Var}(\hat{\theta}) + (\text{bias}(\hat{\theta}))^2,$$

which penalizes both. Unbiasedness is a convenient constraint, not a sacred one.

21.3 Constructing estimators

So far we have *checked* estimators that were pulled out of thin air. Two systematic recipes produce them:

- the **method of moments** (today);
- the **method of maximum likelihood** (Lecture 22).

Definition 21.7 (Population and sample moments). The k -th (**population**) **moment** of a distribution is $\mathbb{E}[X^k]$; the first moment is the mean. The k -th **sample moment** of $X_1, \dots, X_n$ is

$$\overline{X^k} := \frac{X_1^k + X_2^k + \dots + X_n^k}{n}, \quad \text{so in particular} \quad \overline{X^1} = \bar{X}.$$

Definition 21.8 (Method of moments). Let $X_1, \dots, X_n$ be i.i.d. from a distribution with m unknown parameters $\theta_1, \dots, \theta_m$. Set the first m population moments equal to the corresponding sample moments,

$$\mathbb{E}[X^k] = \overline{X^k}, \quad k = 1, 2, \dots, m,$$

and solve the resulting m equations for $\theta_1, \dots, \theta_m$.

The idea in one line:

$$\underbrace{\mathbb{E}[X^k]}_{\text{an expression in } \theta_1, \dots, \theta_m} = \underbrace{\overline{X^k}}_{\text{a number computed from the data}}.$$

The left side is a formula in the unknowns; the right side is something you can evaluate. Write down as many such equations as you have unknowns, then solve. The justification is the law of large numbers: $\overline{X^k} \rightarrow \mathbb{E}[X^k]$ as n grows, so matching them is matching quantities that genuinely become equal.

21.3.1 Gamma

Example 21.9. $X_1, \dots, X_n$ i.i.d. $\text{Gamma}(\alpha, \beta)$, so $m = 2$ unknowns. We know

$$\mathbb{E}[X] = \alpha\beta, \quad \text{Var}(X) = \alpha\beta^2,$$

hence $\mathbb{E}[X^2] = \text{Var}(X) + \mathbb{E}[X]^2 = \alpha\beta^2 + \alpha^2\beta^2$. The two equations are

$$\begin{cases} \bar{X} = \alpha\beta, \\ \overline{X^2} = \alpha\beta^2 + \alpha^2\beta^2. \end{cases}$$

Subtract the square of the first equation from the second. Since $\bar{X}^2 = (\alpha\beta)^2 = \alpha^2\beta^2$, the $\alpha^2\beta^2$ terms cancel:

$$\overline{X^2} - \bar{X}^2 = \alpha\beta^2.$$

Now divide this by the first equation to eliminate α :

$$\frac{\overline{X^2} - \bar{X}^2}{\bar{X}} = \frac{\alpha\beta^2}{\alpha\beta} = \beta,$$

and substitute back into $\bar{X} = \alpha\beta$ to recover α :

$$\hat{\beta} = \frac{\overline{X^2} - \bar{X}^2}{\bar{X}}, \quad \hat{\alpha} = \frac{\bar{X}}{\hat{\beta}} = \frac{\bar{X}^2}{\overline{X^2} - \bar{X}^2}$$

Remark 21.10. Sanity check on the structure: $\overline{X^2} - \bar{X}^2$ is the (biased) sample variance, so the estimators read $\hat{\beta} = \widehat{\text{Var}}/\widehat{\mathbb{E}}$ and $\hat{\alpha} = \widehat{\mathbb{E}^2}/\widehat{\text{Var}}$ — exactly the inversion of $\mathbb{E} = \alpha\beta$, $\text{Var} = \alpha\beta^2$.

With the ten observations 152, 115, 109, 94, 88, 125, 40, 128, 123, 136 you get $\bar{x} = 1110/10 = 111$ and $\overline{x^2} = 132024/10 = 13202.4$, so $\overline{x^2} - \bar{x}^2 = 13202.4 - 12321 = 881.4$ and

$$\hat{\beta} = \frac{881.4}{111} \approx 7.94, \quad \hat{\alpha} = \frac{12321}{881.4} \approx 13.98.$$

21.3.2 Negative binomial

Example 21.11. $X_1, \dots, X_n$ i.i.d. $\text{NegBinomial}(r, p)$, where *both* r and p are unknown. Here

$$\mathbb{E}[X] = \frac{r(1-p)}{p}, \quad \mathbb{E}[X^2] = \frac{r(1-p)(r-rp+1)}{p^2}.$$

Set these equal to $\bar{X}$ and $\overline{X^2}$. As with the gamma, the system is easier after subtracting the square of the first equation from the second, which replaces the second moment by the variance:

$$\bar{X} = \frac{r(1-p)}{p}, \quad \overline{X^2} - \bar{X}^2 = \text{Var}(X) = \frac{r(1-p)}{p^2}.$$

Dividing the first by the second cancels r and $1-p$ outright:

$$\frac{\bar{X}}{\overline{X^2} - \bar{X}^2} = \frac{r(1-p)/p}{r(1-p)/p^2} = p \quad \implies \quad \hat{p} = \frac{\bar{X}}{\overline{X^2} - \bar{X}^2}.$$

Solving the first equation for r gives $r = \bar{X} p / (1-p)$; substituting $1 - \hat{p} = \frac{\overline{X^2} - \bar{X}^2 - \bar{X}}{\overline{X^2} - \bar{X}^2}$ yields

$$\hat{r} = \frac{\bar{X}^2}{\overline{X^2} - \bar{X}^2 - \bar{X}}.$$

Remark 21.12. Note that $\hat{p}$ and $\hat{r}$ are random variables (functions of $X_1, \dots, X_n$); plugging in the observed data turns them into numbers.

21.3.3 Hypergeometric: counting fish

Example 21.13 (Capture–recapture). To estimate the number N of fish in a lake, catch 100, mark them, and release them. A week later catch 80, and suppose 10 of them carry a mark. Estimate N .

The number M of marked fish in the second catch is hypergeometric (Lecture 10): 80 draws without replacement from N fish, 100 of them marked, so $\mathbb{E}[M] = 80 \cdot \frac{100}{N}$. Matching this to the observed 10 is the method of moments with a single equation:

$$80 \cdot \frac{100}{\hat{N}} = 10 \quad \implies \quad \hat{N} = \frac{100 \cdot 80}{10} = 800.$$

In general $\hat{N} = n_1 n_2 / M$, the *Lincoln–Petersen* estimator: the marked fraction of the second catch estimates the marked fraction of the lake.

Remark 21.14. This estimator is biased upward, for a reason worth remembering. Lecture 9 noted that $\mathbb{E}[h(M)] \neq h(\mathbb{E}[M])$ in general; for $h(m) = 1/m$ the inequality always goes the same way, $\mathbb{E}[1/M] \geq 1/\mathbb{E}[M]$, because $1/m$ is convex (this is *Jensen's inequality*). Worse, if $M = 0$ is possible, $\hat{N}$ is infinite with positive probability. A standard fix is Chapman's version $\frac{(n_1+1)(n_2+1)}{M+1} - 1$, here $\frac{101 \cdot 81}{11} - 1 \approx 743$, which is nearly unbiased.

21.4 Where the method of moments runs out

1. **Computing $\mathbb{E}[X^k]$ can be hard.** The method needs closed-form population moments. For the negative binomial above, the second moment was already unpleasant; for many distributions there is no elementary formula at all.
2. **Solving the system can be hard.** With $m \geq 2$ parameters the moment equations are non-linear and need not have a closed-form solution — or any solution.
3. **The answer can fall outside the parameter space.** In the negative binomial, if the sample happens to have variance smaller than its mean, then $\overline{X^2} - \bar{X}^2 - \bar{X} < 0$ and $\hat{r}$ comes out *negative* — impossible, since r counts successes. Nothing in the method prevents this.

Remark 21.15. Method-of-moments estimators are cheap to compute and are a good way to get a starting value, but they carry no guarantee: not of landing in the parameter space, and not of being efficient. Maximum likelihood, next lecture, is built to fix both.